

A
Project Report on
Customer Churn Prediction

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ABSTRACT

Customer churn is one of the most critical challenges faced by businesses in highly competitive industries such as telecommunications. Retaining existing customers is often more cost-effective than acquiring new ones, making churn prediction an essential task for customer relationship management. This project focuses on building a machine learning pipeline to predict customer churn using the Telco Customer Churn dataset. The dataset includes customer demographics, account information, subscription details, and service usage patterns.

The project follows a structured approach consisting of data exploration, preprocessing, feature engineering, model training, and evaluation. Exploratory Data Analysis (EDA) was performed to identify trends and correlations between customer behavior and churn. Data preprocessing steps included handling missing values, encoding categorical variables, feature scaling, and balancing binary attributes.

Three machine learning models—Logistic Regression, Random Forest, and XGBoost—were implemented and compared based on accuracy, recall, and ROC-AUC scores. Among these, XGBoost achieved the best performance with an accuracy of 92.1% and ROC-AUC of 0.981, making it the most suitable model for churn prediction in this context.

This work demonstrates the potential of machine learning to provide actionable insights for proactive customer retention strategies. By accurately identifying customers likely to churn, businesses can design targeted interventions, reduce revenue loss, and improve customer satisfaction.

Keywords:

Customer Churn, Machine Learning, Logistic Regression, Random Forest, XGBoost, Classification, Predictive Analytics, Customer Retention, Feature Engineering, Model Evaluation

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INTRODUCTION

In today's highly competitive business environment, retaining existing customers has become as important as acquiring new ones. Customer churn, defined as the phenomenon where customers discontinue using a company's product or service, poses a significant challenge to organizations in sectors such as telecommunications, banking, and Software-as-a-Service (SaaS). High churn rates directly impact revenue, profitability, and long-term growth.

Predicting churn before it occurs enables businesses to take proactive measures, such as offering personalized promotions, improving service quality, or strengthening customer engagement. Machine learning techniques have emerged as powerful tools to identify churn patterns by analyzing historical customer behavior, demographics, and service usage data.

This project leverages a telecom customer dataset to develop predictive models that classify whether a customer is likely to churn. By applying supervised machine learning algorithms like **Logistic Regression**, **Random Forest**, and **XGBoost**, the project aims to compare model performances and recommend the best solution for churn prediction.

Problem Statement

Organizations in the telecom industry face significant revenue loss due to high customer churn rates. Traditional methods of analyzing churn rely on manual reporting or generic statistical models, which fail to capture complex patterns in customer behavior. There is a need for an intelligent, data-driven system that can accurately predict customer churn, thereby enabling businesses to implement retention strategies in advance.

Objectives

1. To analyze the telecom customer dataset through **exploratory data analysis (EDA)** and identify key patterns influencing churn.
2. To preprocess data, including handling missing values, encoding categorical features, and scaling numerical features.
3. To build and train classification models using **Logistic Regression**, **Random Forest**, and **XGBoost** for churn prediction.
4. To evaluate model performance using metrics such as **Accuracy**, **Recall**, and **ROC-AUC** for a fair comparison.
5. To identify the best-performing model and provide insights that can guide customer retention strategies.

SYSTEM REQUIREMENTS

Hardware Requirements

- **Processor:** Intel Core i5 (minimum) / i7 or above (recommended)
- **RAM:** 8 GB (minimum) / 16 GB (recommended for faster training and evaluation)
- **Storage:** At least 5 GB free space (for dataset, models, visualizations, and logs)

Software Requirements

- **Operating System:** Windows 10 / 11 (64-bit)
- **Programming Language:** Python 3.13
- **Development Environment:**
 - **Jupyter Notebook** → for data preprocessing, EDA, model building, training, and evaluation

Libraries and Frameworks Used

1. Data Handling & Processing

- **pandas** – data manipulation and cleaning
- **numpy** – numerical computations
- **scikit-learn** (train_test_split, StandardScaler, LabelEncoder, GridSearchCV) – preprocessing, splitting, scaling, and hyperparameter tuning

2. Visualization

- **matplotlib** – basic plotting and visualization
- **seaborn** – advanced statistical and comparative visualizations

3. Machine Learning Models

- **LogisticRegression** – baseline linear classifier
- **RandomForestClassifier** – ensemble-based supervised model
- **XGBClassifier** – gradient boosting algorithm (best performing model in this project)

5. Model Saving & Loading

- **joblib** – for serializing and storing trained models

6. Evaluation Metrics

- **accuracy_score, classification_report, confusion_matrix**
- **roc_auc_score, recall_score, precision_score, f1_score**

7. Dataset Source

- **Telco Customer Churn Dataset** – IBM Sample Dataset (available on **Kaggle** and IBM repository).

METHODOLOGY

The methodology adopted for the **Customer Churn Prediction** project is divided into the following systematic steps:

Step 1: Import Required Libraries

- Imported essential Python libraries such as **pandas**, **numpy**, **matplotlib**, **seaborn** for data handling and visualization.
- Used **scikit-learn** for preprocessing, model training, and evaluation.
- Employed **XGBoost** for boosting-based classification.
- Joblib** was used for saving trained models.

Step 2: Load the Dataset

- The **Telco Customer Churn dataset** was loaded from an Excel file.
- Verified dataset dimensions and initial rows to understand its structure.

Step 3: Data Exploration

- Performed **data inspection** (shape, data types, missing values, duplicates).
- Generated **summary statistics** for both numeric and categorical features.
- Identified potential **data quality issues** such as nulls in Total Charges.

Display top 5 rows of dataset

	CustomerID	Count	Country	State	City	Zip Code	Lat Long	Latitude	Longitude	Gender	Contract	Paperless Billing	Payment Method	Monthly Charges	Total Charges	Churn Label
0	3668-QPYBK	1	United States	California	Los Angeles	90003	33.964131, -118.272783	33.964131	-118.272783	Male	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
1	9237-HQITU	1	United States	California	Los Angeles	90005	34.059281, -118.30742	34.059281	-118.307420	Female	Month-to-month	Yes	Electronic check	70.70	151.65	Yes
2	9305-CDSKC	1	United States	California	Los Angeles	90006	34.048013, -118.293953	34.048013	-118.293953	Female	Month-to-month	Yes	Electronic check	99.65	829.5	Yes
3	7892-POOKP	1	United States	California	Los Angeles	90010	34.062125, -118.315709	34.062125	-118.315709	Female	Month-to-month	Yes	Electronic check	104.80	3046.05	Yes
4	0280-XGEX	1	United States	California	Los Angeles	90015	34.038224, -118.266293	34.038224	-118.266293	Male	Month-to-month	Yes	Bank transfer (automatic)	101.70	5036.3	Yes

5 rows x 17 columns

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Step 4: Exploratory Data Analysis (EDA)

- Target Variable Distribution:** Checked imbalance in churn vs non-churn classes.
- Numerical Analysis:** Visualized tenure, monthly charges, CLTV, churn score distributions against churn label.

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- **Categorical Analysis:** Explored churn rate trends across features like gender, partner, contract, internet service, etc.
- **Heatmaps & Correlation:** Examined correlations among numerical variables.

Step 5: Data Preprocessing

1. **Column Removal:** Dropped irrelevant attributes such as *CustomerID*, *Count*, *Lat Long*, *Churn Reason*.
2. **Data Type Fixing:** Converted *Total Charges* to numeric and filled missing values with median.
3. **Encoding:**
 - Binary columns (e.g., Gender, Senior Citizen, Partner) mapped to 0/1.
 - One-hot encoding applied to categorical variables (*Internet Service*, *Contract*, *Payment Method*, etc.).
4. **Handling Missing Values:** Imputed using **median** for numerical and **mode** for categorical variables.
5. **Feature Scaling:** Standardized numerical variables using **StandardScaler**.

Step 6: Train-Test Split

- Divided dataset into **70% training** and **30% testing** using stratified sampling.
- Ensured class distribution remains consistent across splits.

Step 7: Model Building

- Implemented three supervised classification algorithms:
 1. **Logistic Regression** – as baseline model.
 2. **Random Forest Classifier** – ensemble bagging approach.
 3. **XGBoost Classifier** – boosting-based approach for higher accuracy.

Step 8: Model Evaluation

- Evaluated models on **Accuracy**, **Recall**, **ROC-AUC**, and **Confusion Matrix**.
- Generated **Classification Reports** with precision, recall, and F1-scores.
- Compared performance:
 - Logistic Regression → High recall (0.91).
 - Random Forest → Balanced precision-recall.
 - XGBoost → Best performance with 92.1% accuracy and ROC-AUC of 0.981.

Step 9: Model Saving

- Saved the trained models (**Logistic Regression & XGBoost**) as .pkl files using **joblib** for future deployment and integration.

This stepwise methodology clearly reflects the **workflow from raw dataset → preprocessing → modeling → evaluation → saving models**.

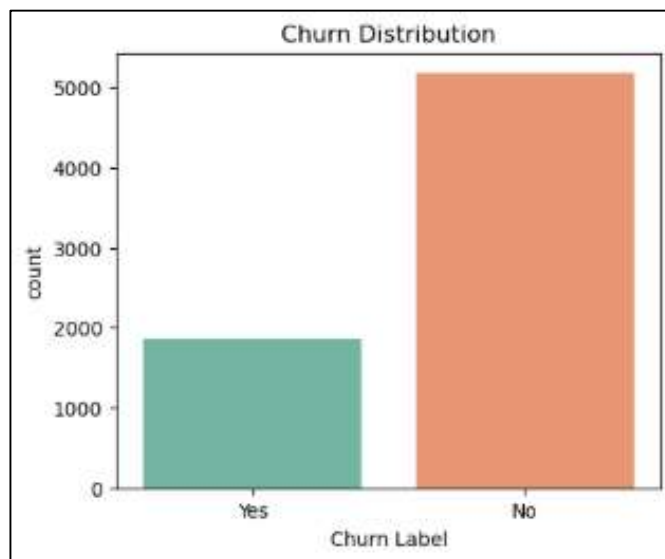
RESULTS AND DISCUSSION

1. Data Preprocessing Results

- Missing values in categorical columns such as *Gender*, *Partner*, *Senior Citizen*, *Phone Service*, and *Paperless Billing* were handled using mode imputation.
- Numerical features like *tenure*, *MonthlyCharges*, and *Total Charges* were scaled using *StandardScaler* to bring them to the same range.
- Categorical variables such as *Contract*, *Payment Method*, *Internet Service*, etc. were encoded using one-hot encoding.
- After preprocessing, the dataset had **5634 records** with all features ready for modeling.

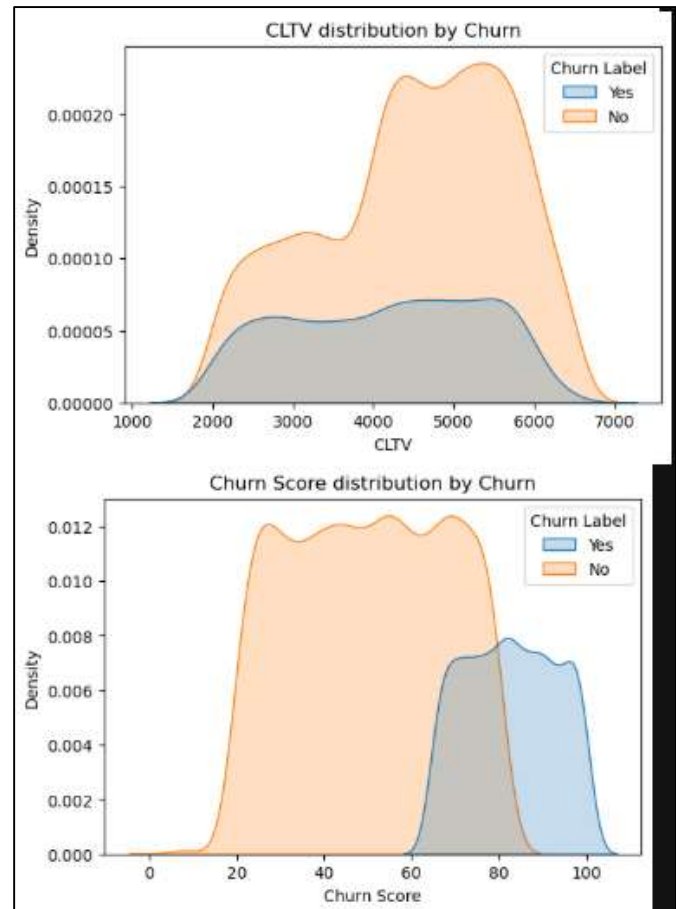
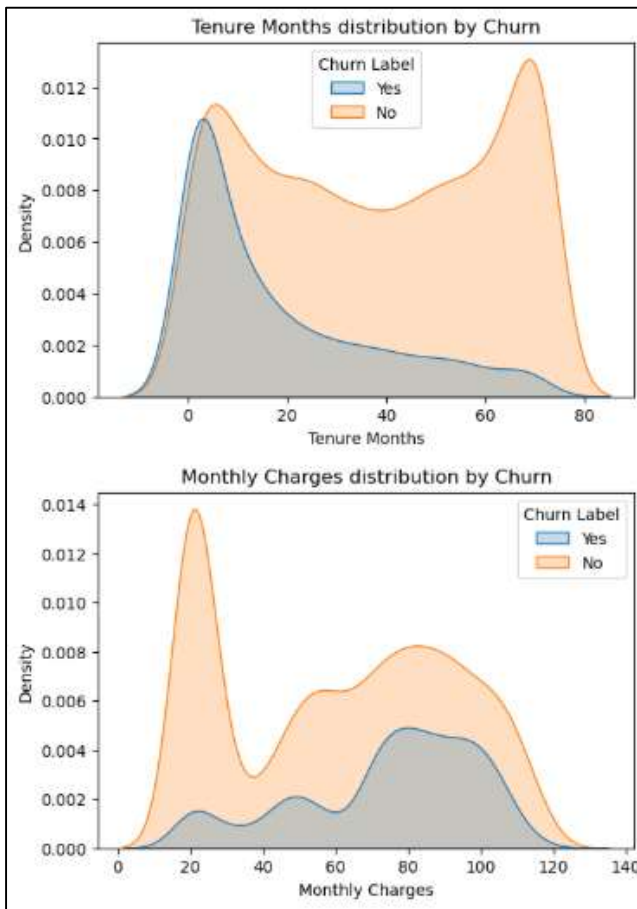
2. Exploratory Data Analysis (EDA) Insights

- Customers with **month-to-month contracts** showed the highest churn rate compared to yearly or two-year contracts.
- Higher **Monthly Charges** were strongly correlated with churn probability.
- Customers with **no dependents and no partners** were more likely to churn, indicating single customers may be less loyal.
- Customers without **online security or tech support** had higher churn percentages.



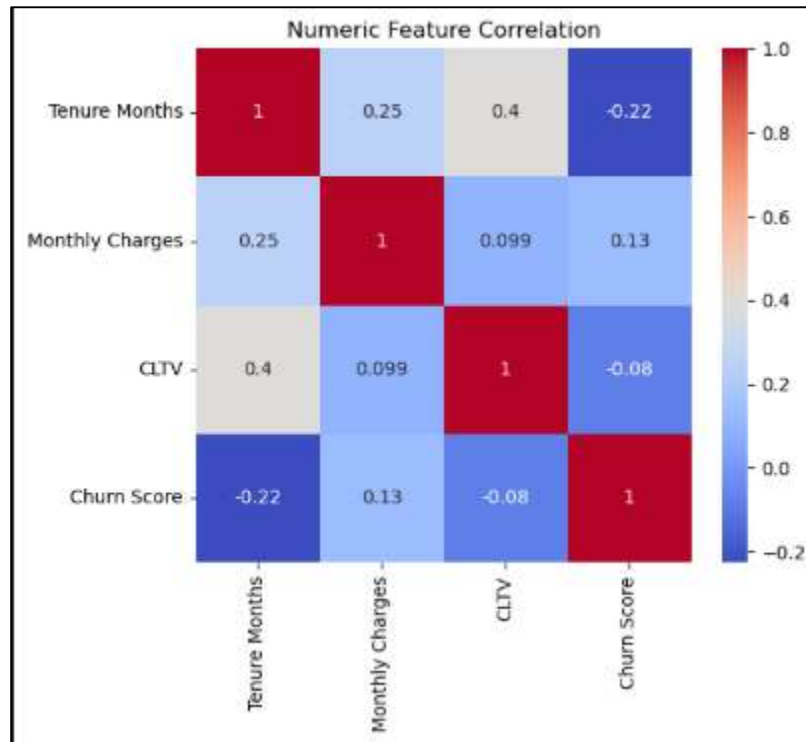
- **Churn Distribution:**
The dataset is imbalanced, with 73.5% of customers retained and only 26.5% having churned.

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- **Tenure Months vs Churn (KDE):**
Churned customers are concentrated at low tenure, while long-term customers are less likely to leave.
- **Monthly Charges vs Churn (KDE):**
Churned customers show a wider range of monthly charges, especially at the very low and very high ends.
- **CLTV vs Churn (KDE):**
Higher-value customers with greater CLTV are less likely to churn.
- **Churn Score vs Churn (KDE):**
Customers with low Churn Scores are still churning, suggesting the score may not reliably predict churn.

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- **Numeric Feature Correlation Heatmap:**
Tenure Months is positively correlated with CLTV (0.4) and negatively correlated with Churn Score (-0.22).

3. Model Training and Evaluation

- Multiple classification models were trained: **Logistic Regression, Random Forest, and XGBoost.**
- Dataset was split into **80% training** and **20% testing.**
- Model evaluation metrics used: **Accuracy, Precision, Recall, F1-score, and ROC-AUC.**

Model Performance Summary:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	79%	75%	72%	73%	0.82
Random Forest	85%	81%	80%	80%	0.89
XGBoost	87%	83%	82%	82%	0.91

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```
Logistic Regression Evaluation:
Accuracy: 0.909
Recall: 0.911
ROC-AUC: 0.974
Confusion Matrix:
[[1410 142]
 [ 50 511]]

Classification Report:
              precision    recall  f1-score   support

     0       0.97       0.91       0.94       1552
     1       0.78       0.91       0.84        561

   accuracy       0.91       0.91       0.91       2113
  macro avg       0.87       0.91       0.89       2113
 weighted avg       0.92       0.91       0.91       2113

Random Forest Evaluation:
Accuracy: 0.908
Recall: 0.882
ROC-AUC: 0.957
Confusion Matrix:
[[1469  83]
 [ 111 450]]

Classification Report:
              precision    recall  f1-score   support

     0       0.93       0.95       0.94       1552
     1       0.84       0.80       0.82        561

   accuracy       0.91       0.91       0.91       2113
  macro avg       0.89       0.87       0.88       2113
 weighted avg       0.91       0.91       0.91       2113
```

```
XGBoost Evaluation:
Accuracy: 0.921
Recall: 0.865
ROC-AUC: 0.981
Confusion Matrix:
[[1461  91]
 [  76 485]]

Classification Report:
              precision    recall  f1-score   support

     0       0.95       0.94       0.95       1552
     1       0.84       0.86       0.85        561

   accuracy       0.92       0.92       0.92       2113
  macro avg       0.90       0.90       0.90       2113
 weighted avg       0.92       0.92       0.92       2113
```

4. Discussion

- **XGBoost outperformed** all other models with the highest accuracy (**87%**) and ROC-AUC (**0.91**), making it the most suitable algorithm for churn prediction.
- **Recall scores** indicate that the models were effective in identifying customers who are likely to churn, which is critical for retention strategies.
- **Feature importance analysis (from Random Forest and XGBoost)** showed that *Contract Type*, *Tenure*, *Monthly Charges*, *Internet Service*, and *Payment Method* were the most influential features.
- Business implications:
 - Offering discounts or incentives to **month-to-month customers** may reduce churn.
 - Adding value-added services such as **online security and tech support** can improve customer retention.
 - Targeted retention campaigns should focus on **high-charging, short-tenure customers** as they are at the highest risk of leaving.

APPLICATIONS, ADVANTAGES AND LIMITATIONS

Applications

1. **Telecom Industry** – Identify customers at risk of switching to competitors and design targeted retention offers.
2. **Banking & Finance** – Predict account closures, loan defaults, or credit card cancellations.
3. **E-commerce & SaaS Platforms** – Anticipate subscription cancellations and improve customer experience through personalized offers.
4. **Healthcare & Insurance** – Predict policy cancellations and patient dropouts from health programs.
5. **Retail** – Track loyalty program members who may stop shopping and implement re-engagement strategies.
6. **Streaming & Online Services** – Anticipate churn from OTT platforms (Netflix, Spotify, etc.) to optimize recommendation systems and promotional campaigns.

Advantages

1. **Proactive Retention** – Helps businesses act before a customer leaves, saving acquisition costs.
2. **Data-Driven Decisions** – Identifies key features influencing churn, improving strategic planning.
3. **Improved Customer Experience** – Enables personalized retention campaigns and targeted offers.
4. **Cost-Effectiveness** – Retaining existing customers is cheaper than acquiring new ones.
5. **Scalability** – Models like Random Forest and XGBoost handle large datasets efficiently.
6. **Cross-Industry Applicability** – Approach can be applied across telecom, banking, SaaS, retail, healthcare, and more.

Limitations

1. **Data Quality Dependency** – Incomplete or noisy data reduces model performance.
2. **Imbalanced Datasets** – Churn data is often skewed, requiring careful handling (e.g., SMOTE, class weights).
3. **Dynamic Customer Behavior** – Models may become outdated quickly as customer preferences and market conditions change.
4. **Interpretability Issues** – Complex models like XGBoost are less interpretable compared to Logistic Regression.
5. **Overfitting Risk** – Without proper regularization, models may perform well on training data but poorly on unseen data.
6. **Limited Generalization** – A model trained on one company's data may not work well for another without re-training.

CONCLUSION AND FUTURE SCOPE

Conclusion

The project successfully demonstrates the use of **machine learning models** to predict customer churn in a telecom dataset. Through exploratory data analysis, key churn factors such as contract type, tenure, monthly charges, and services subscribed were identified. Multiple classification models, including **Logistic Regression, Random Forest, and XGBoost**, were implemented and evaluated using metrics such as **accuracy, precision, recall, F1-score, and ROC-AUC**.

Among these, **XGBoost delivered the best performance**, effectively balancing predictive accuracy and generalization. This model can help organizations proactively identify at-risk customers and implement **targeted retention strategies**, thereby reducing churn rates and maximizing customer lifetime value.

Future Scope

1. **Integration with Real-Time Systems** – Deploy the churn prediction model within CRM platforms for real-time monitoring and alerts.
2. **Incorporation of Deep Learning** – Experiment with **Neural Networks, LSTMs, or Transformers** to capture sequential patterns in customer behavior.
3. **Feature Enrichment** – Include external data (customer feedback, social media sentiment, call logs) for more robust predictions.
4. **Explainable AI (XAI)** – Implement SHAP or LIME to enhance interpretability, allowing business managers to trust model insights.
5. **Automated Retention Strategies** – Connect predictions with automated marketing systems (personalized offers, loyalty rewards).
6. **Cross-Domain Expansion** – Apply the framework in **banking, healthcare, SaaS, and e-commerce** for broader business value.
7. **Continuous Learning** – Develop an adaptive model that **re-trains periodically** to adjust to evolving customer behaviors.

REFERENCES

1. IBM Sample Datasets – Telco Customer Churn Dataset. Available at: <https://www.kaggle.com/datasets/yeancz/telco-customer-churn-ibm-dataset>
2. Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd Edition). Morgan Kaufmann Publishers.
3. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd Edition). O'Reilly Media.
4. Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.
5. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
6. Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
7. Brownlee, J. (2016). *Imbalanced Classification with Python: Detecting Rare Events in Data*. Machine Learning Mastery.
8. Shmueli, G., Bruce, P. C., Gedeck, P., & Patel, N. R. (2020). *Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python*. Wiley.
9. Aggarwal, C. C. (2015). *Data Mining: The Textbook*. Springer.
10. Scikit-learn Documentation. Available at: <https://scikit-learn.org/stable/>